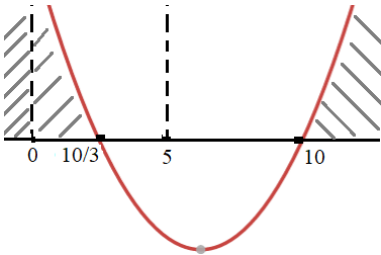
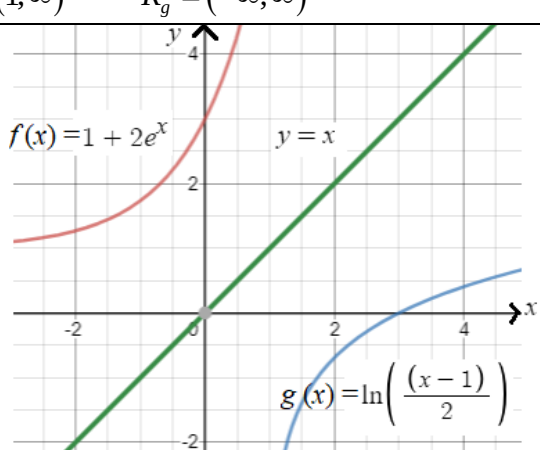


SM015 MATHEMATICS 1
SESSION 2020/2021
ANSWER SCHEME – PKA SET 1

No.	Section A
1a	$\lim_{x \rightarrow -\infty} \sqrt{\frac{3x+1}{6x-3}} = \lim_{x \rightarrow -\infty} \sqrt{\frac{\frac{3x}{x} + \frac{1}{x}}{\frac{6x}{x} - \frac{3}{x}}}$ $= \lim_{x \rightarrow -\infty} \sqrt{\frac{3 + \frac{1}{x}}{6 - \frac{3}{x}}}$ $= \sqrt{\frac{3 + \frac{1}{-\infty}}{6 - \frac{3}{-\infty}}}$ $= \sqrt{\frac{1}{2}} @ \frac{1}{\sqrt{2}} @ \frac{\sqrt{2}}{2} @ 0.707 \text{ (3 sf)}$
1b	$\lim_{x \rightarrow 0} \frac{x-2x^2}{1-\sqrt{x+1}} = \lim_{x \rightarrow 0} \frac{x(1-2x)}{(1-\sqrt{x+1})(1+\sqrt{x+1})} \cdot \frac{(1+\sqrt{x+1})}{(1+\sqrt{x+1})}$ $= \lim_{x \rightarrow 0} \frac{x(1-2x)(1+\sqrt{x+1})}{(1-(x+1))}$ $= \lim_{x \rightarrow 0} \frac{x(1-2x)(1+\sqrt{x+1})}{-x}$ $= \lim_{x \rightarrow 0} \frac{(1-2x)(1+\sqrt{x+1})}{-1}$ $= \frac{(1-2(0))(1+\sqrt{0+1})}{-1}$ $= -2$
2a	$\frac{dy}{dx} = \frac{(x^2-1)(3e^{2x})(2) - (3e^{2x})(2x)}{(x^2-1)^2}$ $= \frac{6e^{2x}(x^2-x-1)}{(x^2-1)^2}$

	$x - 2y = -7 \text{ ————— (2)}$ $(1) + (2): 2x = -8 \Rightarrow x = -4$ $(1) - (2): 4y = 6 \Rightarrow y = \frac{3}{2}$ $z = \frac{3}{2} - 4i$ $ z = \sqrt{\left(\frac{3}{2}\right)^2 + (-4)^2} = \frac{\sqrt{73}}{2}$ $\theta = -\tan^{-1}\left(\frac{4}{3/2}\right) \text{ quadrant 4}^{\text{th}}$ $= -1.21 \text{ rad}$ $\therefore \text{polar form } z = \frac{\sqrt{73}}{2} [\cos(-1.21) + i \sin(-1.21)]$
2a	$\left(\sqrt{2\sqrt{x+1}}\right)^2 = \left(\sqrt{3x-5}\right)^2$ $\left(2\sqrt{x+1}\right)^2 = (3x-5)^2$ $4x+4 = 9x^2 - 30x + 25$ $9x^2 - 34x + 21 = 0$ $(x-3)(9x-7) = 0$ $x = 3, \quad x = \frac{7}{9} \text{ check}$ $\therefore x = 3$
2b	$(11^2)^x (11) \frac{(3)^x}{(3)^2} = \frac{(14)^x}{(14)}$ $\left[\frac{(11^2)^x (3)^x}{(14)^x} \right] = \frac{(3)^2}{(14)(11)}$ $\log \left[\frac{(11^2)(3)}{(14)} \right]^x = \log \left[\frac{(3)^2}{(14)(11)} \right]$ $x = \frac{\log \left[\frac{9}{154} \right]}{\log \left[\frac{363}{14} \right]}$ $x = -0.872 \text{ (3 sf)}$

2c	$\frac{1}{ x-5 } \leq \frac{2}{ x }$ $ x \leq 2 x-5 , \quad x \neq 0, 5$ $(x)^2 \leq (2(x-5))^2$ $x^2 \leq 4x^2 - 40x + 100$ $3x^2 - 40x + 100 \geq 0$ $(3x-10)(x-10) \geq 0$  $(-\infty, 0) \cup \left(0, \frac{10}{3}\right] \cup [10, \infty)$ $\left\{x : x < 0 \cup 0 < x \leq \frac{10}{3} \cup x \geq 10; x \neq 5\right\}$ $\left\{x : x \leq \frac{10}{3} \cup x \geq 10; x \neq 0, 5\right\}$
3a	$f(2) = 7$ $\frac{a(2)+3}{(2)-b} = 7$ $2a + 7b = 11 \text{ ——— (1)}$ $f^{-1}(x) = y$ $f(y) = x$ $\frac{ay+3}{y-b} = x$ $ay + 3 = xy - bx$ $ay - xy = -bx - 3 \Rightarrow y = \frac{-bx-3}{a-x}$ $f^{-1}(x) = \frac{-bx-3}{a-x}$ $f^{-1}(1) = \frac{-b(1)-3}{a-(1)}$ $-4 = \frac{-b-3}{a-1}$

	$4a - b = 7 \text{ ————(2)}$ From (1): $a = \frac{11-7b}{2}$ $4\left[\frac{11-7b}{2}\right] - b = 7$ $-14b = -14 \Rightarrow b = 1$ From (1): $2a + 7(1) = 11 \Rightarrow a = 2$
3b	$(g \circ f^{-1})(c) = g\left[\frac{-c-3}{2-c}\right]$ $\frac{1}{2} = 5 - \left[\frac{-c-3}{2-c}\right]$ $\frac{9}{2} = \frac{-c-3}{2-c}$ $18 - 9c = -2c - 6$ $c = \frac{24}{7}$
4a	let $f(x_1) = f(x_2)$ $1 + 2e^{x_1} = 1 + 2e^{x_2}$ $2e^{x_1} = 2e^{x_2}$ $x_1 = x_2$ Since $x_1 = x_2$, $f(x) = 1 + 2e^x$ is one-to-one function.
4b	$(f \circ g)(x) = f\left[\ln\left(\frac{x-1}{2}\right)\right]$ $= 1 + 2e^{\ln\left(\frac{x-1}{2}\right)}$ $= x$ $(g \circ f)(x) = g(1 + 2e^x)$ $= \ln\left(\frac{1 + 2e^x - 1}{2}\right)$ $= x$ Since $(f \circ g)(x) = (g \circ f)(x) = x$, $f(x)$ and $g(x)$ are inverse of each others.
4c	$D_f = (-\infty, \infty) \quad R_f = (1, \infty)$ $D_g = (1, \infty) \quad R_g = (-\infty, \infty)$
4d	

5a (i)	$\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} 2x + 1$ $= 2(-2) + 1$ $= -3$ $\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} 1 - x^2$ $= 1 - (-2)^2$ $= -3$ $\therefore \lim_{x \rightarrow -2} f(x)$ exist, $\lim_{x \rightarrow -2} f(x) = -3$
5a (ii)	$\lim_{x \rightarrow 0^-} 1 - x^2 = \lim_{x \rightarrow 0^+} 2e^{ax} + b$ $1 - (0)^2 = 2e^{a(0)} + b$ $1 = 2 + b$ $b = -1$ $f'(x) = 2e^{ax}(a) + 0$ $f'(1) = 2e^{a(1)}(a) \qquad f'(0) = 2e^{a(0)}(a)$ $= 2ae^a \qquad = 2a$ $2ae^a = 2a$ $2a(e^a - 1) = 0$ $2a = 0, \quad (e^a - 1) = 0$ $a = 0, \quad e^a = 1$ $a = 0$
5b	$x + 1 = 0$ $x = -1$ $\lim_{x \rightarrow -1^-} \frac{5x^2 + 8x + 4}{x + 1} = -\infty \qquad \lim_{x \rightarrow -1^+} \frac{5x^2 + 8x + 4}{x + 1} = \infty$ The line $x = -1$ is vertical asymptote. $\lim_{x \rightarrow -\infty} \frac{5x^2 + 8x + 4}{x + 1} = \lim_{x \rightarrow -\infty} \frac{\frac{5x^2}{x} + \frac{8x}{x} + \frac{4}{x}}{\frac{x}{x} + \frac{1}{x}}$ $= \lim_{x \rightarrow -\infty} \frac{5x + 8 + \frac{4}{x}}{1 + \frac{1}{x}}$ $= -\infty$ $\lim_{x \rightarrow +\infty} \frac{5x^2 + 8x + 4}{x + 1} = \lim_{x \rightarrow +\infty} \frac{\frac{5x^2}{x} + \frac{8x}{x} + \frac{4}{x}}{\frac{x}{x} + \frac{1}{x}}$ $= \lim_{x \rightarrow +\infty} \frac{5x + 8 + \frac{4}{x}}{1 + \frac{1}{x}}$ $= +\infty$ No horizontal asymptote.
6a	$2x + 2y \frac{dy}{dx} = 2 \frac{dy}{dx}$ $\frac{dy}{dx} - y \frac{dy}{dx} = x$ $\frac{dy}{dx}(1 - y) = x$

	$\frac{dy}{dx} = \frac{x}{1-y}$ $\frac{d^2y}{dx^2} = \frac{(1-y)(1) - x\left(-\frac{dy}{dx}\right)}{(1-y)^2}$ $= \frac{(1-y)(1) + x\left(\frac{x}{1-y}\right)}{(1-y)^2}$ $= \frac{(1-y) + \frac{x^2}{1-y}}{(1-y)^2}$ $= \frac{\frac{(1-y)^2}{(1-y)} + \frac{2y-y^2}{1-y}}{(1-y)^2}$ $= \frac{1-2y+y^2+2y-y^2}{(1-y)} \times \frac{1}{(1-y)^2}$ $\frac{dy}{dx} = \frac{1}{(1-y)^3} \quad \text{Shown!}$ $\left(\frac{2}{x}\right)\left(\frac{x}{1-y}\right)^3 - x^2\left(\frac{1}{(1-y)^3}\right)$ $\frac{2x^2}{(1-y)^3} - \frac{x^2}{(1-y)^3}$ $\frac{x^2}{(1-y)^3}$ $\frac{2y-y^2}{(1-y)^3}$ $\frac{y(2-y)}{(1-y)^3}$
6b (i)	$\frac{dx}{dt} = e^t \text{ and } y = \frac{1}{1+e^t} = (1+e^t)^{-1}$ $\frac{dy}{dt} = -(1+e^t)^{-2}(e^t) = -\frac{e^t}{(1+e^t)^2}$

	$\frac{dy}{dx} = -\frac{e^t}{(1+e^t)^2} \times \frac{1}{e^t}$ $\frac{dy}{dx} = -\frac{1}{(1+e^t)^2}$ since $(1+e^t)^2 > 0$ always positive $-\frac{1}{(1+e^t)^2}$ is always negative $\therefore \frac{dy}{dx} < 0$ Shown!
6b (ii)	$\frac{d^2y}{dx^2} = \frac{d}{dt} \frac{-1}{(1+e^t)^2} \times \frac{1}{e^t}$ $= -(1)(-2)(1+e^t)^{-3}(e^t) \times \frac{1}{e^t}$ $= \frac{2}{(1+e^t)^3}, \quad t=0$ $\frac{d^2y}{dx^2} = \frac{2}{(1+e^0)^3}$ $= \frac{1}{4}$
7b	$\frac{dV}{dt} = 10 \text{ cm}^3\text{s}^{-1}, \quad h = \frac{1}{5}r \rightarrow r = 5h, \quad h = 2$ $V = \frac{1}{3}\pi r^2 h$ $V = \frac{1}{3}\pi(5h)^2 h$ $V = \frac{25}{3}\pi h^3$ $\frac{dV}{dh} = 25\pi h^2$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $= \left[\frac{1}{25\pi h^2} \right] [10]$ $= \left[\frac{1}{25\pi(2)^2} \right] [10]$ $= \frac{1}{10\pi} \text{ cms}^{-1}$